

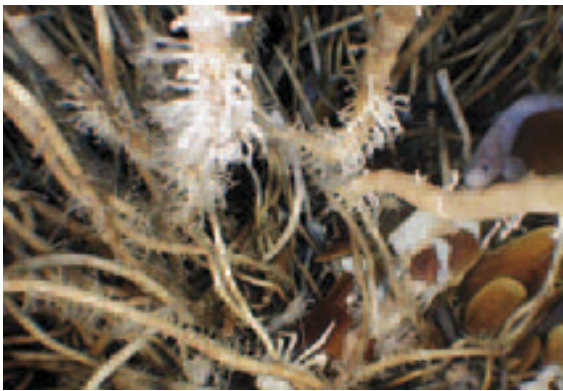


▲ CRABS

The mats of bacteria around Pacific vents are grazed by armies of blind white crabs, and similar swarms of shrimp feed on the bacterial mats around Atlantic vents. Big colonies of mussels and clams also rely on the bacteria for food, but allow it to grow inside their shells, on their gills. Having such a reliable food source enables them to grow fast, and turn into giants with shells up to 10 in (25 cm) long.

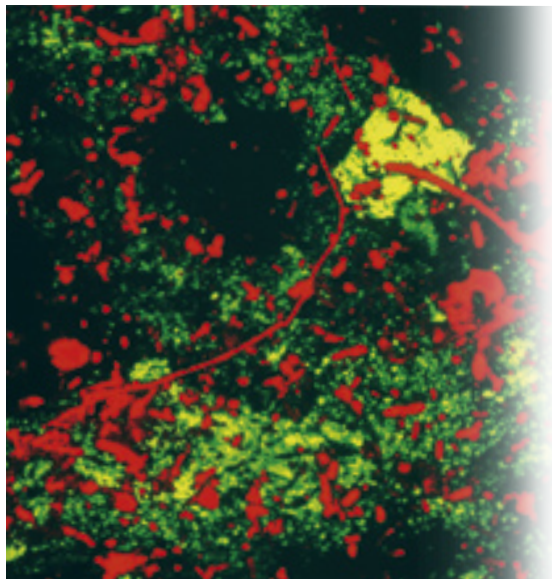
GIANT TUBEWORMS ►

The most spectacular vent animals are giant tubeworms up to 6 ft (2 m) long, with bright red gill plumes. They live in clumps around the vents, where they absorb the chemically rich water and supply it to colonies of bacteria living inside their bodies. The bacteria make food that the worms can use to build their tissues, just like the organisms that live in partnership with tropical reef corals. This enables the worms to grow much faster than most deep-sea life.



▲ METHANE SEEPS

Similar living communities of bacteria have been found around places where methane (natural gas) is seeping from the ocean floor. The high pressure at depth makes the methane freeze into a form of ice. Bacteria process this and use it to make food that supports ice worms and other animals.



◀ LIVING IN THE ROCK

Bacteria that make food by chemosynthesis have been found living beneath the ocean floor, deep in the sediments, and between layers of solid sedimentary rock. Shown here in red, they survive by turning hydrogen and carbon dioxide into methane, producing energy to make their food. Their ability to live in this extreme environment suggests how life may have started on Earth. These organisms may also be able to live in even more extreme conditions—such as beneath the surface of Mars.